

Gene Sight Andro-Male Infertility NGS Panel

Patient name	: Mr. XXX	PIN	: XX
Gender/Age	: Male / XX Years	Sample number	: XX
Reference Clinician	: XX	Sample collection date	: NA
Referring Clinic	: XX	Sample received date	: NA
Specimen	: Peripheral blood	Report date	: NA

INDICATION FOR TESTING

Proband, Mr. XXX semen analysis revealed teratozoospermia. He has undergone 3 cycles of IUI which were unsuccessful and one ART cycle with self-gametes. A total of 8 oocytes were retrieved but there was a total fertilization failure observed even with calcium activation process post ICSI. Proband, Mr. XXX has been evaluated for pathogenic variations and polymorphisms.

RESULTS

MALE POLYMORPHISM ANALYSIS

List of polymorphic variant (SNP) identified:

Gene Name	RsID	Variant	Genotype	Zygosity	Observed Phenotype	Recommendations
NOS3	rs1799983	c.894T>G	GG	Homozygous Polymorphism	Significant reduction in eNOS activity, leading to decreased nitric oxide bioavailability. This may contribute to impaired sperm function and an increased risk of male subfertility [4].	Minimize ROS exposure and handling time in vitro. L-arginine supplementation combined with antioxidants (CoQ10, vitamins C and E) might help improve semen parameters.

MALE FERTILITY SINGLE NUCLEOTIDE VARIATION ANALYSIS

No pathogenic or likely pathogenic variant(s) causative of the reported phenotype was detected

*Genetic test results are based on the recommendation of American college of Medical Genetics [1-3].

No other variant that warrants to be reported for the given clinical indication was identified.

Recommendations

- Genetic counseling is recommended.

Methodology: Single Nucleotide Variation - Single Nucleotide Polymorphism

SNV analysis: DNA extracted from the blood, was used to perform targeted gene capture using a custom capture kit (for SNV analysis – 1109 genes). The targeted libraries were sequenced to a targeted depth of 80 to 100X using SURFSeq 5000 sequencing platform. This kit has deep exonic coverage of all the coding regions including the difficult to cover regions. The sequences obtained are aligned to human reference genome (GRCh38.p13) using Sentieon aligner and analyzed using Sentieon for removing duplicates, recalibration and re-alignment of indels. Sentieon DNAscope has been used to call the variants. Detected variants were annotated and filtered using the VarSeq software with the workflow implementing the ACMG guidelines for variant classification. The variants were annotated using 1000 genomes (V2), gnomAD (v4.1.0,3.1.2,2.1.1), ClinVar, OMIM, dbSNP, NCBI RefSeq Genes. In-silico predictions of the variant was carried out using VS-SIFT, VS-PolyPhen2, PhyloP, GERP++, GeneSplicer, MaxEntScan, NNSplice, PWM Splice Predictor. Only non-synonymous and splice site variants found in the coding regions were used for clinical interpretation. Silent variations that do not result in any change in amino acid in the coding region are not reported.

SNP analysis: Variant analysis (for SNP analysis – 115 genes) and interpretation is done using VarSeq Software. Extensive scientific literature, Information from variant analysis and disease specific databases, population specific research is used to interpret and recommend. All results are finally approved by medical geneticists.

Sequence data attributes

Total reads generated	8.23 Gb
Data ≥ Q30	96.44 %

Genetic test results are reported based on the recommendations of American College of Medical Genetics [1], as described below:

Classification	Interpretation
Pathogenic	A disease-causing variation in a gene which can explain the patients' symptoms has been detected. This usually means that a suspected disorder for which testing had been requested has been confirmed
Likely Pathogenic	A variant which is very likely to contribute to the development of disease however, the scientific evidence is currently insufficient to prove this conclusively. Additional evidence is expected to confirm this assertion of pathogenicity.
Variant of Uncertain Significance	A variant has been detected, but it is difficult to classify it as either pathogenic (disease causing) or benign (non- disease causing) based on current available scientific evidence. Further testing of the patient or family members as recommended by your clinician may be needed. It is probable that their significance can be assessed only with time, subject to availability of scientific evidence.

Disclaimer

- The classification of variants of unknown significance can change over time. Anderson Diagnostics and Labs cannot be held responsible for it.
- Intronic variants, UTR and Promoter region variants are not assessed using this assay.
- Certain genes may not be covered completely, and few mutations could be missed. Variants not detected by this assay may impact the phenotype.
- The variations have not been validated by Sanger sequencing.

- The above findings and result interpretation was done based on the clinical indication provided at the time of reporting.
- It is also possible that a pathogenic variant is present in a gene that was not selected for analysis and/or interpretation in cases where insufficient phenotypic information is available.
- Genes with pseudogenes, paralog genes and genes with low complexity may have decreased sensitivity and specificity of variant detection and interpretation due to inability of the data and analysis tools to unambiguously determine the origin of the sequence data in such regions.
- In addition, this test does not cover detection of such structural variants in the genes *SMN1*, *SMN2*, *HBA1*, *HBA2*, *PMS2*, *GBA* and *CYP21A2* which have very high homology. Moreover, the ORF15 transcript isoform (RefSeq id: NM_001034853.1) of the *RPGR* gene associated with retinitis pigmentosa is not sufficiently covered in the test.
- Incidental or secondary findings that meet the ACMG guidelines can be given upon request [3].
- Result interpretation was done based on the literature evidence available at the time of reporting. The clinical significance of the polymorphic variants tested can change over time and Anderson Diagnostics & Labs cannot be held responsible for this.
- This is not a diagnostic test and so not to be considered as diagnosis of any disease. This test is meant only for understanding the polymorphism at a given position and its association with various clinical parameters.

References

1. Richards, S, et al. Standards and Guidelines for the Interpretation of Sequence Variants: A Joint Consensus Recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. *Genetics in medicine: official journal of the American College of Medical Genetics*. 17.5 (2015): 405-424.
2. Amberger J, Bocchini CA, Scott AF, Hamosh A. McKusick's Online Mendelian Inheritance in Man (OMIM). *Nucleic Acids Res*. 2009 Jan;37(Database issue):D793-6. doi: 10.1093/nar/gkn665. Epub 2008 Oct 8.
3. Lee, Kristy et al. "ACMG SF v3.3 list for reporting of secondary findings in clinical exome and genome sequencing: A policy statement of the American College of Medical Genetics and Genomics (ACMG)." *Genetics in medicine: official journal of the American College of Medical Genetics* vol. 27,8 (2025): 101454. doi:10.1016/j.gim.2025.101454.

4. Chavoshi, Reza, and Seyed Babak Khalifeh Zadeh Kaleybar. "Association of nitric oxidesynthase3 gene rs1799983 G/T polymorphism with idiopathic asthenozoospermia in Iranian Azeri males: a case-control study." Hormone molecular biology and clinical investigation vol. 41,4 10.1515/hmbci-2020-0014. 29 Jun. 2020, doi:10.1515/hmbci-2020-0014.

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